

Do **not** write solutions on this page.

12. [Maximum mark: 20]

Consider the non-zero vectors $\mathbf{u}$ and $\mathbf{v}$. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$.

- (a) Using the definitions of $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ in terms of $|\mathbf{u}|$, $|\mathbf{v}|$ and θ , show that $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2$. [2]

A triangle ABC has vertices $A(0, 1, 2)$, $B(p, q, 3)$ and $C(3, 2, 1)$, $p, q \in \mathbb{Q}$.

The vectors $\mathbf{u}$ and $\mathbf{v}$ are defined as $\mathbf{u} = \vec{AB}$ and $\mathbf{v} = \vec{AC}$.

It is given that $\mathbf{u} \cdot \mathbf{v} = 3$ and the area of triangle ABC is $\sqrt{6}$.

- (b) (i) Find the value of $|\mathbf{u} \times \mathbf{v}|$.
 (ii) Hence, or otherwise, find the value of $|\mathbf{u}|$.
 (iii) Hence, or otherwise, find the possible values of p and the corresponding values of q . [13]

Consider a new point D, the vector $\mathbf{w}$ is defined as $\mathbf{w} = \vec{CD}$.

It is given that $\mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = 0$ and the area of triangle ACD is 5 square units.

- (c) Assuming that $p = 1$, find the possible vectors for $\mathbf{w}$. [5]

